Materials Engineering Reflection

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In the first two weeks of the “Sensing the World” Module, we’ve been given an in-depth look into the fabrication process of chips. After being given an introduction to the basics of chips, we were given the opportunity to create electrodes inside of Whitaker Lab.

Given the equation for resistivity, we were tasked with finding the resistivity of silver paste using only the given materials. After some brainstorming, we as a class came up with the idea that we could mimic the screen-printing process by placing layers of tape on the slide, cutting out a certain size, and then applying the silver paste, which could then be flattened by the rubber squeegee. This process yielded a column of silver paste with known height, width, and depth whose electrical resistance could be measured with a multimeter.

To accomplish this all on time, our group decided to divide the work. One person used a Vernier caliper to find the thickness of the tape, while another began cutting out our electrodes with an X-Acto knife. I measured the length and width of our electrodes using the Vernier caliper. After applying paste, wiping it across with the squeegee, and removing the tape, we moved on to measure the resistance. However, we could not get a reading for resistance. We attributed this to not applying enough silver paste, as we only used one piece of tape and a small area. We improved on this in the next session when we were testing silver paste by using double the amount of tape to ensure there was enough paste to get a reading. We were successful in getting a reading this time and calculated the resistivity, which was relatively close to the actual value for Ag paste.

In our final session at Whitaker Lab, we expanded upon our electrodes by creating a water vapor sensor. We learned about PEDOT-PSS and that when PEDOT-PSS comes in contact with water vapor, it expands and pushes the pathways the electric current travels through apart, leading to a higher resistance reading. We got to see PEDOT-PSS being applied onto our electrode by means of spin coating. Here the centripetal force spread the solution across our glass slide, leaving behind a thin film. We tested it by holding the multimeter onto the electrode while breathing onto it. Across several trials, we consistently saw a momentary increase in resistance. These lab sessions consisted of only the first part of our module, and we spent the rest of the class exploring all that Lehigh has to offer.

Our first tour consisted of Wilbur Powerhouse, where we were introduced to the various types of machinery and tools at our disposal to pursue our own personal projects. A big emphasis of the tour was that everyone was welcome to come and use the space to its full

potential; all you had to do was ask. A couple of highlights of the tour included the vast number of 3D printers available, a CNC machine, and a laser cutter.

Our second tour consisted of the MOCVD and the cleanroom. Here we got to explore photolithography and everything that's involved in the process. We learned the importance of keeping a sterile environment to prevent contaminants such as hair from getting onto the silicon wafers, which leads to patterning defects on them. From here we learned why the room contained yellow lighting. It's to prevent blue light from being present, which would cause the photoresist to react before masking. This ruins the whole point of photolithography, whose goal is to create patterns on the wafer by selecting specific places to apply light.

Lastly, we got to create an electrical circuit using copper tape and LEDs. This gave us a basic understanding of circuit schematics and the notation that comes with them.

Overall, the two weeks gave me a deeper appreciation for the chip fabrication process, and though it seemed complicated at first, the underlying processes all make sense. At the beginning, it was easy to feel overwhelmed with the amount of information, but having the opportunity to apply what we talked about helped to develop my understanding. As we went over the same vocabulary and concepts over and over again, I began to feel more confident about them and the role they played in the chipmaking process. What struck me the most was how intricate the actual chipmaking process is—not only that we're working on the nanometer scale, but the amount of work that goes into creating the machinery that allows you to make the chips themselves. For example, TSMC's lithography machines cost tens of millions of dollars and are big enough to fill an entire room. This just goes to show all the work and engineering that's gone into chipmaking across the years. I've gained a greater appreciation for everyday objects that involve chips, like my laptop and phone.

Something else that surprised me was how chipmaking involves tons of different engineering fields, like Materials, Chemical, and Electrical Engineering, which all must come together not only to create the chip but to be able to replicate it on a mass scale. The process of creating an electrode with only a couple of basic supplies pushed me to think outside the box and change the way I approach problems I've never seen before. From here I learned the most significant part of the overall experience: the importance of collaboration. At times when I or another member of our team struggled, working through it with others helped to take the load off certain members and distribute it across the entire group. This allowed us to work more efficiently as well as providing new perspectives on problems that I or others hadn't thought of before. Just talking through the concepts and hearing them explained in different ways helped me develop my own idea of what they meant in a way that made sense to me. I think the labs were the most helpful part of what

we've done because they provided a visual for largely abstract concepts that are hard to understand just by hearing about them. For example, in the circuit schematic portion, having the antenna of the butterfly complete the circuit and having the LED light up helped me digest what we'd learned and feel more confident with it.